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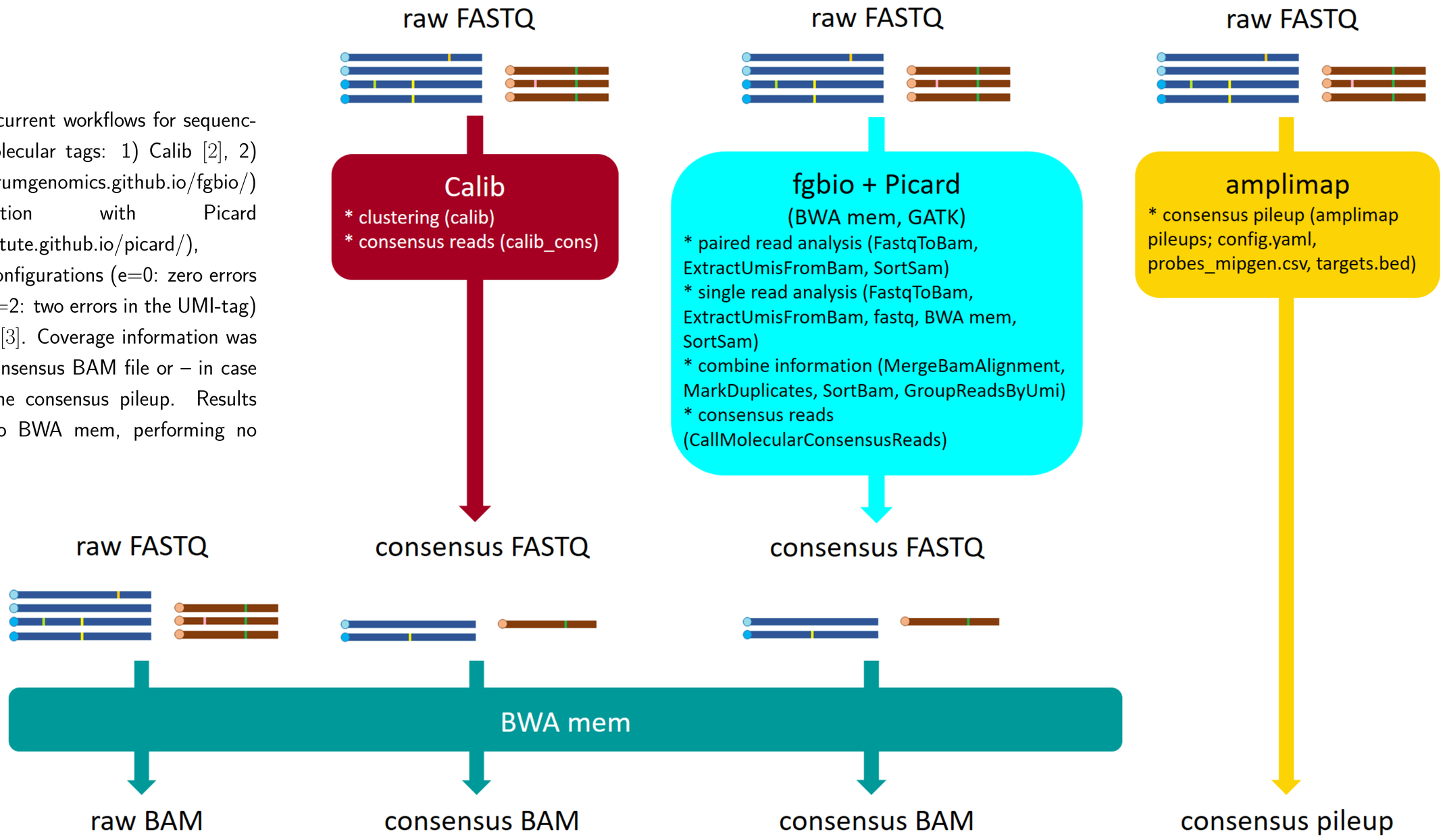
Analyzing coverage in smMIPs data

Background

Common next-generation sequencing (NGS) techniques enable the detection of variants with variant allele frequencies (VAFs) of 1% and above. However, background noise resulting from sequencing artifacts hampers detection of variants with even lower VAFs. Single molecule molecular inversion probes (smMIPs) provide a new technique for detecting variants with VAFs below 1% [1], which play an important role in minimal residual disease as well as clonal evolution. Despite the strength of smMIPs data, a standard analysis workflow generating consensus reads with low background noise, remaining high coverage for subsequent sensitive variant calling, is currently lacking.

Methods

We considered 3 current workflows for sequencing data with molecular tags: 1) Calib [2], 2) fgbio (http://fulcrumgenomics.github.io/fgbio/) in combination with Picard (http://broadinstitute.github.io/picard/), considering two configurations (e=0: zero errors in the UMI-tag; e=2: two errors in the UMI-tag) and 3) amplimap [3]. Coverage information was taken from the consensus BAM file or – in case of amplimap – the consensus pileup. Results were compared to BWA mem, performing no consensus calling.



Results and Discussion

Analyzing 329 samples with anemia or AML, we observe major differences between the different approaches (Table 1). As expected, highest coverage by far can be observed for BWA mem (on average 19,971x). Among tools performing consensus calling, Calib reports highest coverage (3,088x), followed by amplimap (1,379x). Mapping quality is comparable for all approaches – independent of consensus calling. Regarding base quality, consensus calling leads to a general increase. However, while low-quality bases can still be observed for Calib and fgbio+Picard, they are filtered by amplimap. Considering background noise, we expect to observe a major effect resulting from consensus calling. However, the highest level of background noise can be observed for consensus calling with fgbio+Picard (0.76%), while raw data aligned with BWA mem is characterized by 0.62% background noise. Notably lower levels of background noise can be observed for amplimap (0.10%) and Calib (0.28%). Considering the fact that we aim at variant calling with VAF<1%, only amplimap and Calib appear to provide suitable workflows for smMIPs data that allow detection of variants with <1% VAF.

Table 1: Data characteristics for the raw BAM files, consensus BAM files and consensus pileup files considering coverage, mapping quality, base quality and background noise evaluating 329 test samples.

		coverage					mapping quality			base quality			background noise	
		mean	per sample [min-max]	per base [min-max]	cov=0 in all samples bp	[%]	mean	per sample [min-max]	per base [min-max]	mean	per sample [min-max]	per base [min-max]	mean	per sample [min-max]
BWA mem	raw BAM	19,971	[195-267,397]	[0-85,793]	0	0.00%	59.54	[57.52-59.63]	[18.14-60.00]	35.13	[34.53-35.29]	[10.68-37.00]	0.62%	[0.51%-0.99%]
Calib	consensus BAM	3,089	[131-27,058]	[0-25,121]	0	0.00%	59.07	[46.74-59.82]	[16.39-60.00]	41.66	[41.37-41.78]	[11.29-42.00]	0.28%	[0.17%-1.51%]
fgbio+Picard (e=0)	consensus BAM	181	[6-838]	[0-615]	64	0.21%	59.52	[59.38-59.78]	[17.99-60.00]	44.02	[43.67-44.75]	[13.31-45.00]	0.76%	[0.17%-1.08%]
fgbio+Picard (e=2)	consensus BAM	52	[6-178]	[0-161]	64	0.21%	59.52	[59.38-59.78]	[17.99-60.00]	44.02	[43.69-44.75]	[12.00-45.00]	0.76%	[0.17%-1.07%]
amplimap	consensus pileup	1,379	[2-28,938]	[0-4,720]	72	0.24%	/	/	/	37.00	[36.98-37.00]	[33.80-39.11]	0.10%	[0.06%-0.58%]

References

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